

Savitzky-Golay optimizer diagnostics

Objective: mean squared translational/angular acceleration residual; angular residual uses the reference inertia/mass metric

Tolerances: {'ftol': None, 'xtol': None, 'gtol': 1e-06, 'note': 'ftol and xtol are disabled by default in the SG estimator; gtol or max_nfev terminates the solve unless explicitly overridden'}

Selected strict-ZOH solve

status=1 success=True nfev=6 njev=6

message='gtol' termination condition is satisfied.

reported optimality=1.9842485698973178e-07

initial={'cost': 861.2680039625353, 'gradient_inf_norm': 0.11928309591885977, 'gradient_l2_norm': 0.18748127070906595, 'jacobian': {'condition_number_nonzero_subspace': 428.48050253055806, 'numerical_rank': 14, 'numerical_tolerance': 2.1551452232734554e-10, 'singular_values': [226.77362673980463, 214.29426550356115, 175.24845107644475, 48.58599758279531, 44.6676151421513, 25.86141014039502, 13.13258698129094, 12.188036693929684, 11.740454815333408, 10.944347801319344, 9.770147614088302, 3.8234876787260434, 3.778475568698358, 0.5292507486350134]}, 'residual_l2_norm': 41.50344573556599}

final={'cost': 861.2679776106611, 'gradient_inf_norm': 1.9842485698973178e-07, 'gradient_l2_norm': 2.1903596418071417e-07, 'jacobian': {'condition_number_nonzero_subspace': 429.11118502627755, 'numerical_rank': 14, 'numerical_tolerance': 2.1551401467436814e-10, 'singular_values': [226.77309256556168, 214.29560128444058, 175.25024268427092, 48.586483486776665, 44.666594599569855, 25.860799326756986, 13.132552711639162, 12.18874632026363, 11.741436738839491, 10.943448058869993, 9.769881364487219, 3.823201153520987, 3.7787167244667663, 0.528471642032996]}, 'residual_l2_norm': 41.50344510063378}

coordinate_step_l2_norm=0.0012366333467694983

cost_reduction=2.63518742258384e-05

gradient_tolerance_satisfied=True

finite_difference_max_relative_error=4.050542697931485e-12

finite_difference_directional_checks=[{'analytic_l2_norm': 187.27042613044492, 'difference_l2_norm': 7.58546857101222e-10, 'direction': 'negative_gradient', 'finite_difference_l2_norm': 187.27042613045248, 'relative_error': 4.050542697931485e-12, 'scheme': 'central', 'step': 6.0554544523933395e-06}]

All smoothstep and strict-ZOH stage details are in optimizer_diagnostics.json.